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KENYA

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FIRST REVIEWS UNDER THE TWENTY-FOUR MONTH STAND-BY ARRANGEMENT AND THE ARRANGEMENT UNDER THE STANDBY CREDIT FACILITY AND REQUESTS FOR WAIVERS OF APPLICABILITY, REPHASING OF DISBURSEMENTS, AND MODIFICATION OF PERFORMANCE CRITERION—DEBT SUSTAINABILITY ANALYSIS UPDATE

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Risk of external debt distress:	Low
Augmented by significant risks stemming from domestic public and/or private external debt?	No

Kenya's risk of external debt distress remains low, while overall public sector debt dynamics continue to be sustainable.[1] However, margins have generally narrowed, and standardized stress tests suggest that the vulnerability to export shocks has increased. Public debt has risen in recent years, with most new debt financing infrastructure to address bottlenecks and boost sustainable growth. The bulk of Kenya's external public debt carries concessional terms, but recent commercial borrowing entails significant repayment needs in 2017 (2015 syndicated loan), in 2019 and, especially, in 2024 (2014 sovereign bond issuance). Implementation of the planned reduction in the fiscal deficit over the medium term is essential to limit and eventually reverse the rise in public debt ratios. In addition, the composition of fiscal financing between domestic and foreign sources should seek to contain risks of public external debt from export shocks while avoiding a crowding out of domestic bank credit to the private sector.

[1] Kenya's policies and institutions are classified as "strong" under the World Bank's Country Policy and Institutional Assessment (CPIA) Index (average score in 2013–15: 3.83). The relevant indicative thresholds for this category are: 50 percent for the NPV of debt-to-GDP ratio, 200 percent for the NPV of debt-to-exports ratio, 300 percent for the NPV of debt-to-revenue ratio, 25 percent for the debt service-to-exports ratio, and 22 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

BACKGROUND

1. This DSA consists of two parts: external and public. The external DSA covers external debt of the central government and the central bank, as well as of the private sector; stress tests apply to public and publicly guaranteed (PPG) debt. The public DSA covers total debt—external and domestic—incurred or guaranteed by the central government. Public domestic debt comprises central government debt. Government finance statistics are to be expanded to cover the recently created county governments. In this analysis, total public debt refers to the sum of public domestic and public external debt, but does not cover the entire public sector (e.g., parastatal borrowing without a government guarantee is not covered).

2. Kenya's overall public debt has increased in the past two years. Gross public debt increased from 44 percent of GDP at end-2013 to 52.4 percent at end-2015. In the baseline, public debt is expected to stabilize around 54–55 percent of GDP in 2017–18 and gradually decline thereafter. Half of Kenya's public debt is owed to external creditors.

3. Most of Kenya's external public debt remains on concessional terms, but its commercial component has increased. Nominal public external debt at end-2015 was US$16.1 billion (text table), equivalent to 26.4 percent of GDP.

- Multilateral creditors account for just under half of external credit to Kenya as the share of debt from commercial creditors has increased. Debt to bilateral creditors is roughly half to Paris Club creditors, and half to other bilateral creditors, mainly semi-concessional loans from China to finance construction of the first phase (Mombasa-Nairobi) of the Standard Gauge Railway project (SGR). In December 2015, Kenya contracted a further semi-concessional loan from China of US$1.5 billion, to be disbursed in the coming years, for the second SGR phase (Nairobi-Naivasha).

- Kenya's commercial financing has two main elements. In 2014, Kenya issued its inaugural sovereign bonds, at 5-year and 10-year maturities, raising US$2 billion in June and a further US$750 million in December 2014.2 More recently, in October 2015, Kenya contracted a two-year US$750 million syndicated loan at LIBOR plus 570 basis points, equivalent to an effective yield of 8 percent.

4. Kenya's gross domestic public debt was 26 percent of GDP at end-2015. Domestic debt is issued mostly in the form of Treasury bonds (66 percent of domestic debt) and Treasury bills (32 percent). Commercial banks hold 57 percent of the domestic debt, with nonbanks (pension and trust funds, insurance companies and others) holding 43 percent. Rollover risks appear moderate. The average maturity of Kenya's domestic debt shortened, from 5.8 years in June 2011 to 5.2 years in June 2013 and 4.3 years in June 2016.

[2] The June 2014 issuance comprised two tranches: a five-year $500 million bond at a yield of 5.875 percent, and a 10-year $1.5 billion bond at 6.875 percent. In December 2014, Kenya added $250 million to the five-year tranche at a 5.0 percent yield and $500 million to the 10-year tranche at 5.9 percent.

Kenya External Public Debt

	2012		2013		2014		2015	
	Billion US$	Share	Billion US$	Share	Billion US$	Share	Billion US$	Share
Mulilateral creditors	5.56	57.6	6.47	60.5	7.07	49.2	7.30	44.3
Bilateral creditors	2.71	28.1	2.84	26.6	3.86	26.9	5.18	31.4
Commercial creditors	0.68	7.0	0.69	6.5	2.82	19.6	3.60	21.8
Others (supplier credits)	0.18	1.9	0.18	1.7	0.18	1.3	0.00	0.0
Total (excluding guarantees)	9.13	94.6	10.18	95.2	13.93	96.9	16.08	97.5
Publicly guaranteed debt	0.52	5.4	0.51	4.8	0.44	3.1	0.41	2.5
Total (including guarantees)	9.65	100.0	10.69	100.0	14.37	100.0	16.49	100.0

Source: Kenyan National Treasury.

UNDERLYING ASSUMPTIONS

5. This DSA is based on macroeconomic assumptions that are consistent with the framework for the accompanying staff report. Compared with the previous projections, the growth outlook is modestly weaker and external commercial borrowing is lower in light of a slower pace of public investment.

- Real GDP growth is modestly weaker in the short term, reflecting mainly the expected impact of interest rate caps, but recovers in the medium term. Faster growth in the longer term is predicated on the assumption that the present infrastructure push successfully addresses key bottlenecks.

- The primary fiscal deficit peaked in 2015 owing to a frontloading of SGR-related spending, and is projected to be brought down in the medium to longer term consistent with the East African Community (EAC) Monetary Union convergence criteria.3 The higher primary balance compared with the previous DSA mainly reflects delayed SGR spending in FY2015/16 that spilled over to FY2017/18. In the long term, the primary deficit would converge to its previously projected level, consistent with keeping the overall deficit below 3 percent while making room for higher interest payments.

- The current account deficit narrowed substantially in 2015 owing mainly to lower imports of oil products and investment-related imports. Exports were also weaker than expected due to temporary regulatory issues with the EU in the horticulture sector and lower tourism following the terrorist attacks. The current account is projected to narrow further in 2016 on account of lower oil prices and as exports recover from temporary shocks. In the medium term fiscal consolidation as well as a pickup in trade is expected as demand in key export markets accelerates, progress is made in regional integration, in part due to infrastructure investment, and reforms to the business environment contribute to gains in competitiveness.[4]

[3] The EAC Monetary Union Protocol provides for fiscal convergence criteria, including a ceiling on the fiscal deficit (defined including grants) of 3 percent of GDP; and a ceiling on the gross public debt of 50 percent of GDP in present value terms. The other macroeconomic convergence criteria include a ceiling on headline inflation (8 percent) and a floor on reserve cover (4.5 months of imports). The fiscal plans outlined in the authorities' budget policy statement are consistent with Kenya meeting the EAC convergence criteria by 2021.

[4] Kenya's ranking in the World Bank's Doing Business Report (2016) improved 21 places relative to 2015, one of the top 10 increases across the globe. Ongoing reforms will, amongst other areas, seek to further reduce transportation and trading costs.

The macro-framework projections do not include potential exports from recent oil discoveries. Both exports and imports are lower than previously projected, as a share of GDP.

Kenya: Selected Macroeconomic Assumptions

	2015	2016	2017	2018	2019	Long-term 1/
Real GDP Growth						
Current DSA	5.6	6.0	5.3	5.8	6.2	6.5
Previous DSA (March 2016)	5.6	6.0	6.1	6.5	6.5	6.5
Primary Fiscal Deficit (percent of GDP)						
Current DSA	4.9	4.1	3.3	1.5	0.4	0.2
Previous DSA (March 2016)	5.4	4.1	2.6	1.3	0.7	0.6
Non-interest Current Account (percent of GDP)						
Current DSA	5.8	4.4	4.5	4.3	4.2	4.2
Previous DSA (March 2016)	7.4	7.0	5.5	5.4	5.4	5.9

Source: IMF staff estimates.
1/ For current DSA average 2022–36. For previous DSA, average 2021–35.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

6. External debt indicators remain well below the policy-dependent debt burden thresholds under the DSA baseline scenario. The PV of PPG external debt as a share of GDP is expected to rise moderately to 22.6 percent in 2017 from 20.3 percent in 2015. This is lower than projections in the previous DSA reflecting a lower rate of externally financed investment projects. The PV of PPG external debt then gradually declines over time in line with the planned fiscal consolidation, remaining well below the 50 percent indicative threshold. The PV of external debt to exports and revenues both remain well below the indicative thresholds of 200 percent and 300 percent respectively.

7. Under the standard stress tests, all relevant indicators remain below the thresholds with the exception of one temporary breach.

- Under most extreme stress scenario of an exports shock, the PV of debt to exports breaches the threshold for two years by an average of 8.0 percentage points (Figure 1 and Table 2). This is related to temporary shocks in 2015 which lowered services and horticulture exports resulting in a lower export base. Looking forward, services exports have already begun to recover due to the removal of travel restrictions reflecting substantial improvements in counter-terrorism strategies, which have resulted in a decline in security and crime-related incidents. Horticulture exports have also recovered as the horticulture sector in Kenya has established systems to ensure EU regulatory standards are met on a consistent basis. Moreover, the discovery of significant oil and gas resources—which is an important upside to GDP growth, fiscal balances, and export projections—have not been incorporated into the baseline.

- A one-time nominal exchange rate depreciation of 30 percent is the shock which has the largest impact on the PV of debt to GDP and debt to revenue ratios. Under both of these scenarios, debt remains well below the indicative thresholds.

- Shocks to the debt service indicators relative to exports and revenues also remain well below the relevant thresholds, with peaks reflecting repayment of the 2014 sovereign bond.

8. Staff continues to assess that the risk of external debt distress remains low, although risks have increased. The breach in the threshold of the PV of external debt to exports ratio is small and temporary, and therefore staff continues to assess that the risk of external debt distress is low. The increase in public investment, in part financed by external commercial borrowing, has led to an increase in vulnerabilities. The authorities intend to maximize reliance on concessional borrowing to keep financing costs low and limit refinancing risks. In addition, continued commitment to fiscal consolidation will be key to maintaining debt sustainability. Risks to export growth remain, particularly from security-related threats. If export growth fails to materialize, this could adversely impact Kenya's risk rating (highlighted in Figure 1, panel c).

PUBLIC DEBT SUSTAINABILITY ANALYSIS

9. Total public debt as a share of GDP is set to stabilize in 2017–18 and then gradually decline over the medium term. In 2016, public sector debt is projected to rise to 54.4 percent, from 52.4 percent in 2015. Looking forward, fiscal consolidation will reduce the primary deficit to 1.5 percent of GDP in 2018 down to 0.4 percent of GDP in 2019 resulting in a decline in debt to 52.9 percent of GDP. In PV terms, the debt is stable at 48-49 percent of GDP from 2016-2018 and falls thereafter. The authorities' medium-term debt anchor is 45 percent in PV terms (within the EAC convergence criterion of 50 percent). The PV of public debt-to-revenue ratio would ease from 241 percent in 2016 to 209 percent in 2020 (Table 4).

10. Total public debt as a share of GDP remains well below the LIC DSA public debt benchmark. Under the baseline and all stress scenarios, debt remains below the benchmark of 74 percent of GDP in PV terms. This is the benchmark applicable for LICs whose CPIA score for quality of policies and institutions is assessed as strong and covers the entire public sector. In Kenya, public debt does not include legacy debts of the pre-devolution county governments (whose size is not yet fully clear) as well as borrowings of state-owned enterprises. In addition, public debt should include planned annuities intended to finance road construction: although the annuity obligations may not necessarily be classified as debt under local law, they nevertheless represent public debt obligations according to international (GFS) methodology.

11. The alternative scenarios and bound tests indicate that the projected paths for public debt indicators remain within the relevant benchmarks. Under a standard scenario that keeps the primary balance unchanged from its 2016 level, the PV of public debt to GDP would remain on a steady upwards trajectory but does not exceed the 74 percent benchmark during the projection period. There are risks to this outlook linked to a slippage in the authorities' medium-term fiscal consolidation plans, and possible recourse to commercial external borrowing. In addition, fixing the primary deficit at the 2016 level results in a gradual increase in the debt to revenue ratio, rising to 254 percent in 2020.

MAIN FINDINGS AND CONCLUSIONS

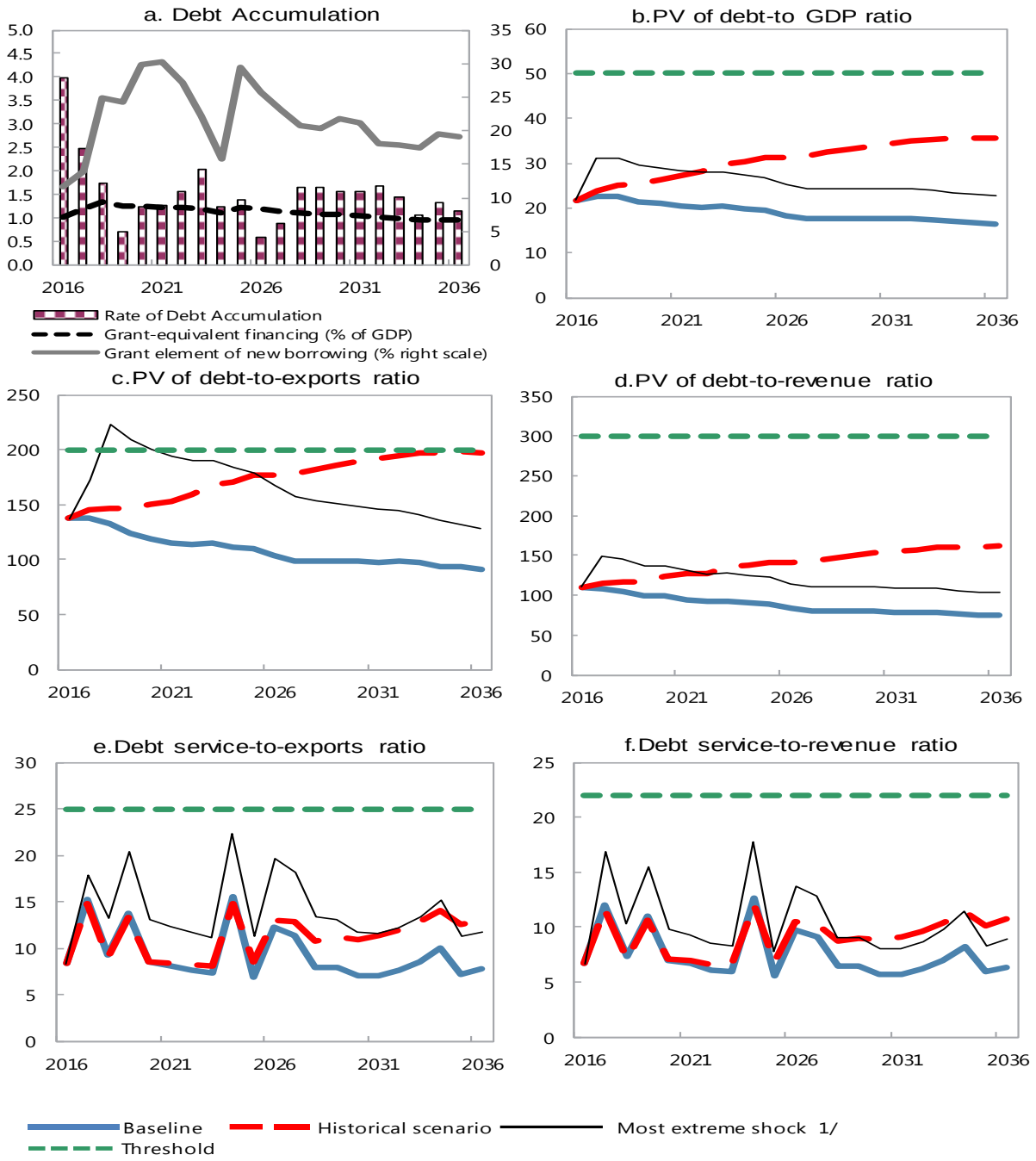
12. This DSA finds that Kenya continues to face low risk of external debt distress. External debt under the baseline and standard stress tests scenarios show that debt remains within sustainable bounds. Standardized stress tests indicate a small and temporary breach under the exports shock. A large exchange rate shock continues to represent a significant risk to external debt. The authorities are strengthening their debt management capacity to manage and prepare for large repayments on commercial borrowing. At the same time, concessional borrowing will continue to play an important role in financing investment projects

due to its lower cost and longer maturity profile. Looking ahead, the planned investment in infrastructure will be critical to raise growth and export potential, both of which will support Kenya's external debt sustainability.

13. Fiscal consolidation is essential to ensure debt remains sustainable. The increase in debt is due to a much needed boost to public investment. Fiscal consolidation began last year and the authorities will need to remain committed to their medium-term plans of gradual adjustment. The baseline public debt path remains consistent with the EAC convergence criteria (deficit and debt) and debt indicators are below the relevant public debt benchmark in the baseline and stress scenarios, subject to coverage issues. Standard stress-testing scenarios show that if the primary deficit were to remain at current levels, public debt would remain on an upward path which highlights the need to follow through on the intended medium-term fiscal consolidation.

14. The authorities are committed to their public investment plan to raise growth in output and exports while ensuring debt sustainability. The increase in infrastructure is continuing and the authorities are committed to fiscal consolidation to maintain debt sustainability. They agree that they remain at low risk of external debt distress. However, they emphasize that their externally financed infrastructure investments will address bottlenecks that have constrained past growth and export performance.

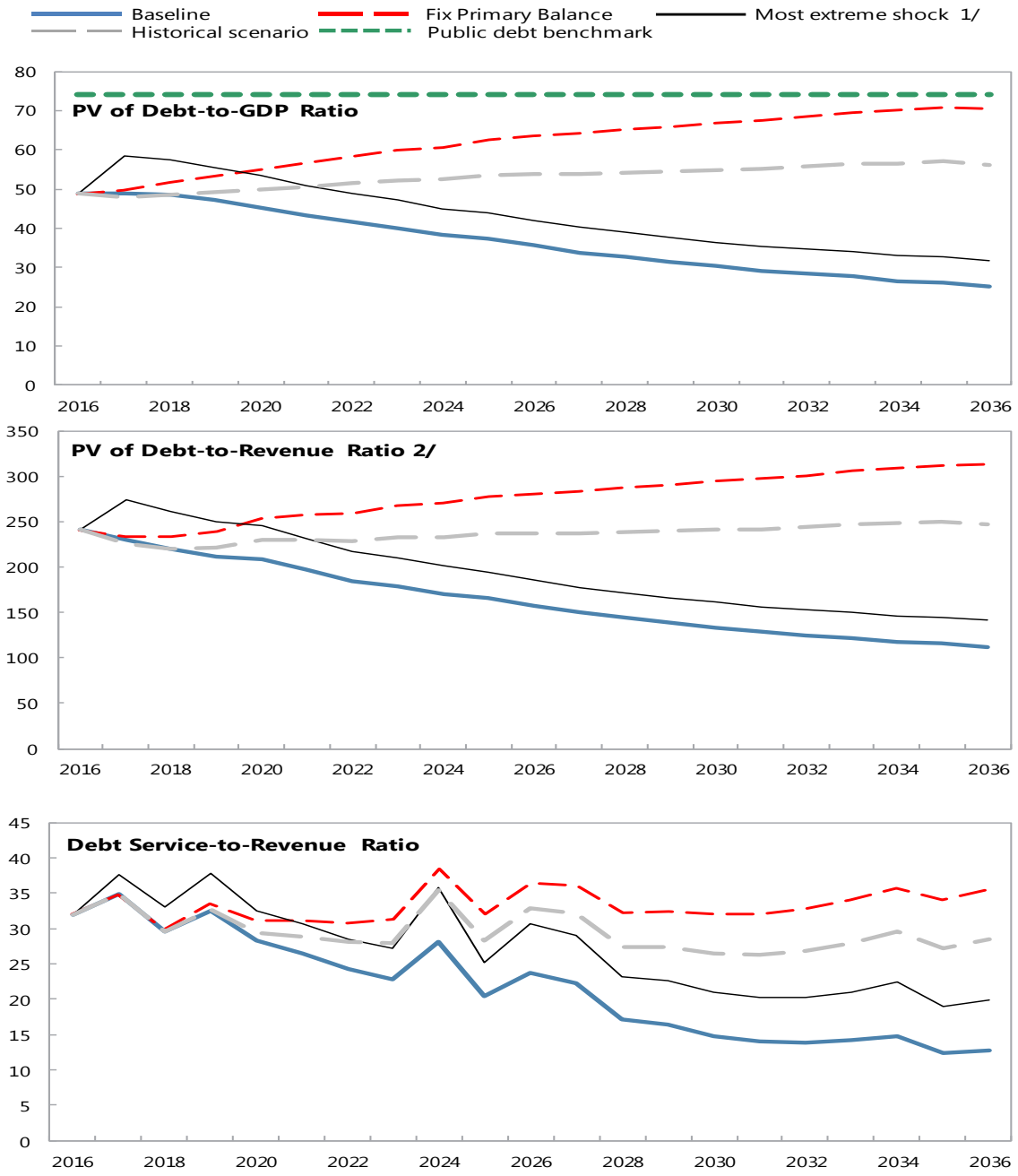
Figure 1. Kenya: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Kenya: Indicators of Public Debt Under Alternatives Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.

Table 1. Kenya: External Debt Sustainability Framework, Baseline Scenario, 2013–36 1/

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	**29.4**	**36.6**	**42.6**			**42.9**	**44.9**	**47.0**	**46.7**	**46.7**	**46.9**		**51.7**	**61.7**	
of which: public and publicly guaranteed (PPG)	19.4	23.5	26.4			27.4	28.3	28.3	27.2	26.8	26.2		23.9	20.8	
Change in external debt	1.0	7.3	6.0			0.3	2.0	2.1	-0.4	0.0	0.2		0.8	1.0	
Identified net debt-creating flows	4.7	5.0	3.9			1.3	1.6	0.9	0.5	0.3	0.4		0.7	0.9	
Non-interest current account deficit	**7.8**	**9.1**	**5.8**	**5.8**	**2.6**	**4.4**	**4.5**	**4.3**	**4.2**	**4.2**	**4.2**		**4.3**	**4.2**	4.2
Deficit in balance of goods and services	13.6	15.1	11.4			11.2	10.3	10.2	9.7	9.7	9.7		9.4	9.5	
Exports	19.9	18.4	16.5			15.8	16.4	17.0	17.3	17.6	17.7		17.6	18.0	
Imports	33.5	33.5	27.9			27.0	26.7	27.2	27.0	27.3	27.4		27.1	27.5	
Net current transfers (negative = inflow)	-5.6	-5.7	-5.4	-6.1	0.5	-4.8	-4.5	-4.4	-4.4	-4.3	-4.2		-3.8	-3.8	-3.9
of which: official	-0.2	-0.1	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	-0.2	-0.3	-0.2			-2.0	-1.3	-1.5	-1.2	-1.2	-1.2		-1.3	-1.5	
Net FDI (negative = inflow)	**-1.7**	**-1.7**	**-1.7**	**-1.4**	**1.1**	**-1.9**	**-2.1**	**-2.3**	**-2.3**	**-2.4**	**-2.3**		**-2.0**	**-1.4**	-1.8
Endogenous debt dynamics 2/	**-1.4**	**-2.4**	**-0.2**			**-1.2**	**-0.8**	**-1.1**	**-1.3**	**-1.5**	**-1.5**		**-1.6**	**-1.9**	
Contribution from nominal interest rate	1.0	0.7	1.0			1.1	1.3	1.3	1.3	1.3	1.3		1.4	1.7	
Contribution from real GDP growth	-1.5	-1.4	-2.0			-2.4	-2.1	-2.4	-2.7	-2.8	-2.8		-3.0	-3.6	
Contribution from price and exchange rate changes	-0.9	-1.6	0.8			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.7**	**2.3**	**2.1**			**-1.0**	**0.4**	**1.3**	**-0.9**	**-0.3**	**-0.2**		**0.1**	**0.1**	
of which: exceptional financing	-0.6	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		...	...	
PV of external debt 4/	...	...	36.5			37.2	39.2	41.2	40.9	40.9	41.0		46.0	57.2	
In percent of exports	...	...	220.8			235.6	239.2	242.8	236.7	232.8	231.9		261.0	317.0	
PV of PPG external debt	**...**	**...**	**20.3**			**21.7**	**22.6**	**22.5**	**21.4**	**20.9**	**20.3**		**18.3**	**16.3**	
In percent of exports	**...**	**...**	**122.8**			**137.4**	**137.9**	**132.2**	**124.1**	**119.3**	**115.0**		**103.5**	**90.6**	
In percent of government revenues	**...**	**...**	**107.7**			**110.1**	**108.7**	**104.7**	**98.9**	**98.9**	**94.8**		**82.8**	**74.4**	
Debt service-to-exports ratio (in percent)	**52.7**	**58.4**	**55.4**			**29.7**	**45.8**	**44.2**	**41.4**	**34.8**	**36.7**		**53.9**	**56.0**	
PPG debt service-to-exports ratio (in percent)	**4.4**	**16.3**	**7.3**			**8.3**	**15.2**	**9.2**	**13.8**	**8.4**	**8.0**		**12.2**	**7.7**	
PPG debt service-to-revenue ratio (in percent)	**4.5**	**15.5**	**6.4**			**6.7**	**12.0**	**7.3**	**11.0**	**7.0**	**6.6**		**9.8**	**6.3**	
Total gross financing need (Billions of U.S. dollars)	11.1	13.2	11.8			10.7	13.5	15.0	17.8	18.6	21.0		45.2	143.2	
Non-interest current account deficit that stabilizes debt ratio	6.8	1.8	-0.2			4.1	2.4	2.1	4.6	4.1	4.0		3.5	3.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.7	5.3	5.6	5.2	2.2	6.0	5.3	5.8	6.2	6.5	6.5	6.0	6.5	6.6	6.5
GDP deflator in US dollar terms (change in percent)	3.5	5.9	-2.1	6.5	7.7	2.2	3.5	1.7	2.3	1.7	2.1	2.2	2.6	2.6	2.6
Effective interest rate (percent) 5/	3.9	2.6	2.9	2.7	0.9	2.9	3.2	3.2	3.1	3.1	3.1	3.1	3.0	3.1	3.1
Growth of exports of G&S (US dollar terms, in percent)	-2.0	2.9	-7.0	7.6	11.3	3.6	13.0	11.5	10.4	10.1	9.5	9.7	9.1	9.6	9.4
Growth of imports of G&S (US dollar terms, in percent)	3.0	11.5	-13.7	11.0	13.8	4.7	7.8	9.5	7.8	9.5	8.9	8.0	8.9	9.6	9.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	11.7	14.0	24.9	24.2	29.9	30.3	22.5	25.7	19.2	21.3
Government revenues (excluding grants, in percent of GDP)	19.2	19.3	18.8			19.7	20.8	21.5	21.7	21.2	21.5		22.1	22.0	22.1
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			1.1	1.3	1.4	1.4	1.5	1.6		2.4	4.8	
of which: Grants	0.3	0.3	0.3			0.3	0.4	0.5	0.5	0.5	0.5		0.9	2.1	
of which: Concessional loans	...	...	...			0.8	0.9	0.9	0.9	1.0	1.1		1.6	2.6	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.0	1.2	1.3	1.3	1.2	1.2		1.2	0.9	1.1
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			20.4	24.2	38.3	38.3	43.2	43.1		39.2	35.9	35.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	55.1	61.5	63.6			68.9	75.1	80.7	87.7	95.0	103.2		160.1	389.0	
Nominal dollar GDP growth	9.3	11.5	3.5			8.3	9.0	7.5	8.6	8.3	8.7	8.4	9.3	9.3	9.2
PV of PPG external debt (in Billions of US dollars)			12.3			14.9	16.6	17.9	18.5	19.6	20.7		28.9	62.8	
(PVt-PVt-1)/GDPt-1 (in percent)						4.0	2.5	1.8	0.7	1.2	1.3	1.9	0.6	1.1	1.4
Gross workers' remittances (Billions of US dollars)	1.3	1.4	1.6			1.8	2.0	2.1	2.3	2.4	2.6		3.8	9.7	
PV of PPG external debt (in percent of GDP + remittances)	...	...	19.8			21.1	22.0	21.9	20.9	20.4	19.8		17.8	16.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	106.8			117.7	119.0	114.7	108.0	104.1	100.6		91.3	79.6	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	6.4			7.1	13.1	8.0	12.0	7.4	7.0		10.8	6.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36

(Percent)

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	22	23	22	21	21	20	**18**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	22	24	25	25	26	27	**31**	36
A2. New public sector loans on less favorable terms in 2016-2036 2	22	23	24	23	23	23	**24**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	22	23	23	22	22	21	**19**	17
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	22	24	28	27	26	25	**22**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	22	23	24	23	22	22	**19**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	22	23	24	22	22	21	**19**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	22	24	27	26	25	24	**21**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	22	31	31	30	29	28	**25**	23
PV of debt-to-exports ratio								
Baseline	137	138	132	124	119	115	**104**	91
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	137	145	147	147	150	153	**177**	197
A2. New public sector loans on less favorable terms in 2016-2036 2	137	141	141	135	133	132	**135**	140
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	137	135	130	122	117	113	**102**	89
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	137	173	223	209	200	193	**167**	128
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	137	135	130	122	117	113	**102**	89
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	137	139	139	130	125	120	**107**	90
B5. Combination of B1-B4 using one-half standard deviation shocks	137	159	188	176	168	163	**142**	112
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	137	135	130	122	117	113	**102**	89
PV of debt-to-revenue ratio								
Baseline	110	109	105	99	99	95	**83**	74
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	110	114	116	117	124	126	**142**	162
A2. New public sector loans on less favorable terms in 2016-2036 2	110	111	111	108	111	109	**108**	115
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	110	109	108	102	102	98	**85**	77
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	110	116	130	122	122	117	**98**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	110	111	111	105	105	101	**88**	79
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	110	109	110	103	103	99	**86**	74
B5. Combination of B1-B4 using one-half standard deviation shocks	110	115	126	118	118	114	**96**	78
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	110	150	145	137	137	132	**115**	103

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	8	15	9	14	8	8	**12**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	8	15	9	13	9	8	**13**	13
A2. New public sector loans on less favorable terms in 2016-2036 2	8	15	8	13	8	8	**9**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	8	15	9	14	8	8	**12**	8
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	8	18	13	20	13	12	**20**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	8	15	9	14	8	8	**12**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	8	15	9	14	9	8	**13**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	8	17	12	18	11	11	**17**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	8	15	9	14	8	8	**12**	8
Debt service-to-revenue ratio								
Baseline	7	12	7	11	7	7	**10**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	7	12	7	11	7	7	**10**	11
A2. New public sector loans on less favorable terms in 2016-2036 2	7	12	7	11	7	7	**7**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	7	12	8	12	7	7	**10**	7
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	7	12	8	12	8	8	**12**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	7	13	8	12	8	7	**11**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	7	12	7	11	7	7	**10**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	7	12	8	12	8	7	**11**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	7	17	10	15	10	9	**14**	9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	17	17	17	17	17	17	**17**	17

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Kenya: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–36

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	44.0	48.6	52.4			54.4	54.7	54.4	52.9	51.2	49.1		41.3	29.7	
of which: foreign-currency denominated	19.4	23.5	26.4			27.4	28.3	28.3	27.2	26.8	26.2		23.9	20.8	
Change in public sector debt	2.3	4.6	3.8			2.0	0.3	-0.3	-1.5	-1.8	-2.1		-1.7	-1.2	
Identified debt-creating flows	1.5	2.9	4.1			1.4	1.9	-0.2	-1.4	-1.9	-2.3		-1.9	-0.2	
Primary deficit	3.0	4.7	4.9	2.2	1.7	4.1	3.3	1.5	0.4	-0.1	-0.3	1.5	0.0	1.2	0.2
Revenue and grants	19.7	19.8	19.3			20.2	21.4	22.1	22.3	21.7	22.0		22.6	22.5	
of which: grants	0.5	0.5	0.5			0.5	0.6	0.6	0.6	0.5	0.5		0.5	0.5	
Primary (noninterest) expenditure	22.7	24.4	24.2			24.3	24.6	23.6	22.7	21.6	21.7		22.6	23.7	
Automatic debt dynamics	-1.5	-1.7	-0.8			-2.7	-1.4	-1.7	-1.8	-1.8	-1.9		-1.9	-1.4	
Contribution from interest rate/growth differential	-0.9	-1.5	-1.8			-1.5	-1.4	-1.6	-1.9	-1.9	-1.8		-1.6	-1.2	
of which: contribution from average real interest rate	1.3	0.7	0.8			1.5	1.3	1.4	1.3	1.3	1.3		1.0	0.7	
of which: contribution from real GDP growth	-2.2	-2.2	-2.6			-3.0	-2.7	-3.0	-3.2	-3.2	-3.1		-2.6	-1.9	
Contribution from real exchange rate depreciation	-0.6	-0.2	1.1			-1.2	0.0	-0.1	0.1	0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.8	1.7	-0.3			0.6	-1.6	-0.1	-0.1	0.2	0.2		0.2	-0.9	
Other Sustainability Indicators															
PV of public sector debt	...	...	46.3			48.7	49.0	48.6	47.1	45.3	43.2		35.6	25.2	
of which: foreign-currency denominated	...	...	20.3			21.7	22.6	22.5	21.4	20.9	20.3		18.3	16.3	
of which: external	...	...	20.3			21.7	22.6	22.5	21.4	20.9	20.3		18.3	16.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	14.8	18.9	17.5			17.5	18.0	15.2	14.6	12.9	12.0		10.1	6.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	239.9			241.1	229.2	220.1	211.5	208.7	196.6		157.5	112.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	245.7			247.2	235.7	226.6	217.4	214.1	201.4		161.4	114.9	
of which: external 3/	...	...	107.7			110.1	108.7	104.7	98.9	98.9	94.8		82.8	74.4	
Debt service-to-revenue and grants ratio (in percent) 4/	29.1	39.2	31.6			32.0	34.8	29.6	32.5	28.4	26.5		23.7	12.8	
Debt service-to-revenue ratio (in percent) 4/	29.9	40.2	32.4			32.8	35.8	30.5	33.4	29.1	27.1		24.3	13.1	
Primary deficit that stabilizes the debt-to-GDP ratio	0.8	0.0	1.1			2.1	3.0	1.8	1.9	1.7	1.8		1.6	2.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.7	5.3	5.6	5.2	2.2	6.0	5.3	5.8	6.2	6.5	6.5	6.0	6.5	6.6	6.5
Average nominal interest rate on forex debt (in percent)	1.4	2.4	2.6	1.6	0.5	2.9	3.2	3.2	3.1	3.1	3.1	3.1	2.9	3.2	3.0
Average real interest rate on domestic debt (in percent)	6.3	2.6	1.7	1.4	4.0	4.4	3.8	4.5	4.3	4.5	4.5	4.3	4.0	3.8	4.1
Real exchange rate depreciation (in percent, + indicates depreciation	-3.2	-1.2	4.7	-3.0	8.0	-4.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.4	8.0	9.1	9.1	3.6	6.0	5.8	5.4	5.5	5.3	5.1	5.5	5.1	5.1	5.1
Growth of real primary spending (deflated by GDP deflator, in percer	12.1	13.2	4.6	3.0	5.3	6.4	6.8	1.5	1.9	1.4	6.8	4.1	6.7	8.1	7.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	11.7	14.0	24.9	24.2	29.9	30.3	22.5	25.7	19.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Refers to gross debt of the central government, including CBK obligations to the IMF, excluding government deposits.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Kenya: Sensitivity Analysis for Key Indicators of Public Debt, 2016–36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	49	49	49	47	45	43	36	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	49	48	49	49	50	51	54	56
A2. Primary balance is unchanged from 2016	49	50	52	53	55	57	63	70
A3. Permanently lower GDP growth 1/	49	49	49	48	47	45	42	43
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	49	50	52	52	51	49	45	39
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	49	50	51	50	48	46	37	26
B3. Combination of B1-B2 using one half standard deviation shocks	49	49	51	50	49	47	41	33
B4. One-time 30 percent real depreciation in 2017	49	59	58	56	53	51	42	32
B5. 10 percent of GDP increase in other debt-creating flows in 2017	49	58	57	56	53	51	42	29
PV of Debt-to-Revenue Ratio 2/								
Baseline	241	229	220	211	209	197	157	112
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	241	225	220	221	230	230	237	247
A2. Primary balance is unchanged from 2016	241	233	234	239	254	257	281	312
A3. Permanently lower GDP growth 1/	241	231	223	217	216	207	184	191
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	241	236	237	232	233	224	199	175
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	241	232	232	223	220	207	166	117
B3. Combination of B1-B2 using one half standard deviation shocks	241	231	231	224	224	213	182	147
B4. One-time 30 percent real depreciation in 2017	241	274	261	250	246	231	186	141
B5. 10 percent of GDP increase in other debt-creating flows in 2017	241	272	260	249	246	232	185	129
Debt Service-to-Revenue Ratio 2/								
Baseline	32	35	30	32	28	26	24	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	32	35	29	33	29	29	33	28
A2. Primary balance is unchanged from 2016	32	35	30	34	31	31	36	35
A3. Permanently lower GDP growth 1/	32	35	30	33	29	27	27	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	32	36	31	35	31	29	28	20
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	32	35	30	33	30	28	25	13
B3. Combination of B1-B2 using one half standard deviation shocks	32	35	30	34	30	28	26	17
B4. One-time 30 percent real depreciation in 2017	32	38	33	38	32	31	31	20
B5. 10 percent of GDP increase in other debt-creating flows in 2017	32	35	32	37	33	31	27	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.